

# VGADASH: Failure-Time Observability Through a VGA-Resident Kernel Dashboard

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**Abstract:** Kernel debugging becomes difficult precisely when it is most needed. During early boot failures, graphics stack regressions, failed userspace initialization, or network outages, a Linux machine may still be executing kernel code while becoming effectively unreachable to the developer. In those states, common observability tools such as `journalctl -k`, SSH-based diagnostics, or GUI consoles are either unavailable or unreliable. VGADASH addresses this problem by providing a small in-kernel dashboard rendered directly in VGA text mode. The system is designed as a local fallback visibility path that exposes recent kernel log activity and a compact summary of system state without depending on normal userspace interaction. Its primary operator interface is the keyboard through SysRq, allowing the dashboard to be shown or navigated even when shell access is impractical [1]. The contribution is not a general observability framework, but a lightweight, recovery-focused mechanism for machines that are alive yet difficult to debug. Validation is grounded in a reproducible QEMU-based study and supporting snapshot checks that provide deterministic evidence for the visible behavior [7].

**Keywords:** Linux kernel module; failure-time observability; VGA text mode; SysRq; printk; QEMU

## 1. Introduction

Systems that fail during kernel experimentation often do not fail cleanly. A machine may stop short of a usable login shell, never complete the graphics handoff, lose remote network reachability, or hang while probing a newly modified driver. In all of these situations, the kernel may still be alive enough to execute some code, but the developer loses the normal paths used to observe what the system is doing. This mismatch creates a practical debugging problem: the system is still running, but it is no longer practically inspectable.

The usual Linux observability stack assumes a healthy dependency chain. Kernel messages are emitted through `printk`, retained in kernel-managed buffers, and later surfaced through tools such as `dmesg` or `journalctl -k` for consumption through a shell, GUI, storage device, or remote session [2]. If userspace initialization fails, if graphics is broken, or if network and storage paths are unavailable, the operator may lose access to information even though the information still exists inside the kernel. In other words, observability fails downstream of the components that are most fragile during early and low-level failures.

VGADASH is proposed as a local fallback observability mechanism for those failure-time conditions. Instead of treating visibility as a userspace service, it treats visibility as a kernel-resident capability. The system renders a small dashboard directly onto the VGA text display path and is intended to be activated through SysRq so that the operator can reach it using the keyboard and monitor alone [1]. The dashboard provides two focused views: a recent-kernel-logs page and a compact state page. This keeps the design small, direct, and aligned with the kind of information that is most useful during degraded recovery scenarios.

This paper asks a focused research question: can a kernel-resident VGA dashboard preserve useful failure-time observability on an alive-but-unusable system with fewer dependencies than a userspace logging stack? The answer is explored through a prototype that reduces the dependency chain to a local display and a keyboard-triggered control path.

This paper makes three contributions. First, it frames failure-time observability as a dependency-reduction problem and motivates a local kernel-resident solution. Second, it presents the design of VGADASH as an integrated combination of VGA text rendering, in-kernel log capture, and SysRq-driven activation. Third, it evaluates the system in a reproducible QEMU-based setting, showing that the design can be studied in a controlled manner [7].

## 2. Problem Context and Motivation

The paper is motivated by a common pattern in kernel and low-level systems work: the more experimental the environment becomes, the less reliable the normal debugging path is. Early boot hangs can occur before a usable shell

appears. Driver changes can break suspend and resume or hang during probe. Network-based diagnostics disappear when NIC initialization or userspace networking fails. Graphics failures can leave a machine without a usable display stack even while the kernel continues to execute. In each case, the machine may be technically alive but operationally opaque.

This opacity is the core problem. Standard observability tools are not wrong; they are simply positioned too high in the dependency stack for the failure mode being considered. `journalctl -k`, for example, is useful under normal conditions, but it presumes healthy userspace services and a path back to the operator. Remote logging solutions shift the dependency to the network stack and a receiver [3]. Post-mortem tools such as `kdump` and `pstore` are valuable after crashes, but they serve a different diagnostic goal [4][5]. VGADASH therefore targets a narrower but important space: systems that have not fully crashed, but are no longer comfortable to debug through conventional interfaces.

From this viewpoint, the problem is best understood as a dependency minimization exercise. If the purpose is to recover some visibility in degraded states, then the display path should be as primitive and local as possible, the control path should not depend on a shell, and the information path should avoid fragile userspace services. That motivation directly leads to the choices at the heart of VGADASH: VGA text mode for display, SysRq for control, and in-kernel capture of recent logging activity. This emphasis on preserving a small working path through failure also aligns with broader systems thinking that reduces reliance on complex recovery chains [8].

### 3. Background

Three background mechanisms matter for understanding the system. The first is `printk`, the Linux kernel logging interface. Kernel subsystems and drivers emit messages through `printk`, and these messages are retained in a kernel-managed buffer before being exposed to higher-level tools [2]. This means that kernel-originated information already exists inside kernel space before any userspace logger displays or persists it. That observation is the foundation of VGADASH's log design.

The second background mechanism is VGA text mode. Although modern graphics systems are far more sophisticated, VGA text mode remains a simple and direct output path on many x86-oriented virtualized environments. It is attractive for a fallback dashboard precisely because it is primitive: it requires neither a userspace compositor nor a complex rendering stack. A small amount of information can therefore be displayed in a deterministic 80x25 text layout when richer interfaces are unavailable.

The third background mechanism is SysRq. The Magic SysRq interface already exists in Linux as an emergency keyboard control path for system recovery and debugging [1]. It is not a general user interface, but that is exactly why it is useful here. SysRq offers a realistic way to invoke a kernel-level recovery surface with minimal dependence on normal shell interaction. In the context of VGADASH, SysRq is not a side feature; it is the primary operator entrypoint.

### 4. System Design

VGADASH combines three coordinated paths: a display path, an information path, and a control path. The display path is a non-destructive VGA text overlay. When activated, the system saves the current text buffer, renders the dashboard, and preserves a restore path so that normal console contents can be returned. This matters because the goal is not to replace the system console permanently, but to provide a temporary diagnostic surface that can be invoked and dismissed as needed.

The information path is intentionally narrow. VGADASH exposes a logs page and a state page. The logs page is meant to answer the immediate question, "what has the kernel been saying recently?" The state page is meant to answer, "what sort of machine state am I looking at right now?" This split is small enough to keep the dashboard readable in text mode, while still giving the operator two distinct types of diagnostic value: event stream visibility and quick situational context.

The control path is built around SysRq. The current interaction model supports a toggle key and direct page-selection keys for the logs and state pages [1]. This choice reflects the paper's core use case. If the machine is degraded, relying on a shell or filesystem-visible control node weakens the argument. SysRq keeps the interface close to the keyboard and closer to the kernel's own emergency control mechanisms.

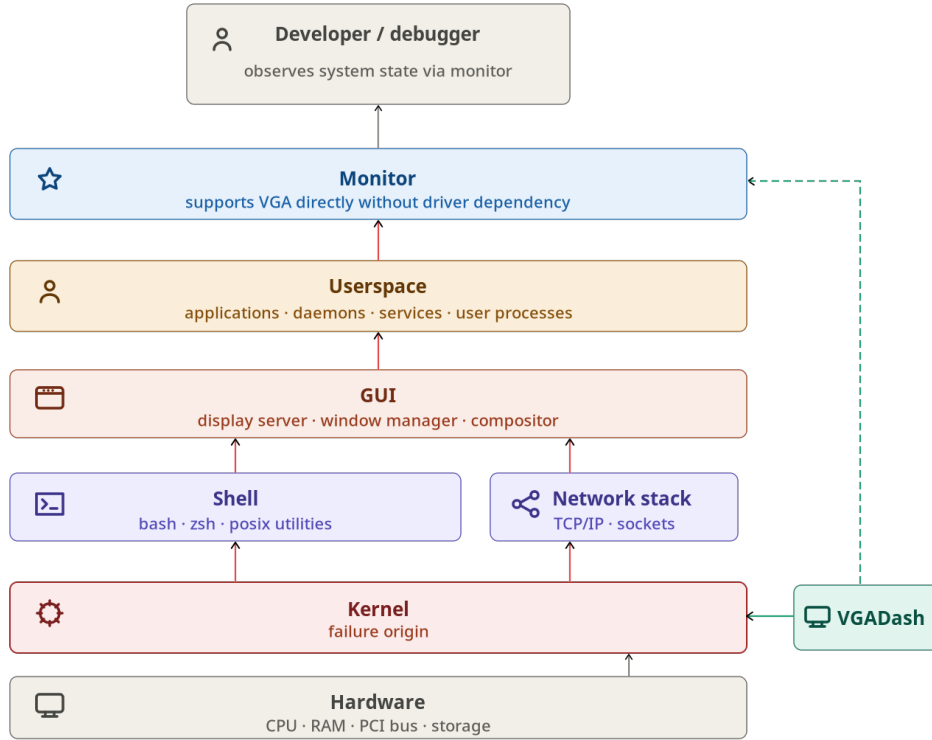


Figure 1. Failure-time observability gap and how VGADASH shortens the dependency chain.

Two design properties are especially important. First, the overlay is non-destructive. It does not claim ownership of the console forever; it temporarily uses the screen and provides a restore path. Second, the interaction model is deliberately small. VGADASH is not intended to become a general debugging shell. It is meant to preserve a thin but valuable observability layer for a machine that is no longer conveniently debuggable.

## 5. Evaluation

The current evaluation is intentionally focused on operational scenarios rather than on performance microbenchmarks. The strongest claim made by this paper is that VGADASH is useful for systems that are still alive but have become difficult to inspect through normal userspace channels. This claim is examined in a controlled QEMU-based setting in which the dashboard is present before userspace becomes relevant, can be reached through SysRq, and can be checked against expected diagnostic output [7].

### 5.1 Methodology and Setup

Experiments were run in a reproducible QEMU environment using an initramfs that places VGADASH in the boot path before a normal interactive session would be required. The kernel command line keeps the system in a minimal text-console configuration, and SysRq is used to invoke the dashboard pages. For supporting checks, a known log marker is introduced into the kernel message path and the logs page is verified against that expected output. This setup focuses on validating the intended failure-time operating path rather than measuring performance.

Three results matter most. First, the logs page provides immediate visibility into recent kernel activity without relying on normal userspace log consumption as the primary narrative path. Second, SysRq-based invocation makes the dashboard reachable in a keyboard-driven recovery path that is much closer to the failure-time use case than a shell-first interface [1]. Third, the QEMU-based evaluation can be repeated consistently enough to support confidence in the observed behavior rather than relying on a single visual example.

The evaluation is therefore best understood as scenario validation. It shows that a machine with broken or unavailable higher-level interfaces can still expose useful information locally, that a keyboard-triggered emergency path is practical for page activation and navigation, and that the visual design can be supported by deterministic tests using exported text snapshots. Figure 2 shows the logs page reached through the SysRq path, while Figure 3 shows the complementary state page used for quick situational context.

```

UGADASH [page:logs]
-----
Last console-emitted kernel log lines (post-load):
[ 1.988827] Run /init as init process
[ 1.988930]   with arguments:
[ 1.988980]   /init
[ 1.989012]   with environment:
[ 1.989042]   HOME=/
[ 1.989068]   TERM=linux
[ 2.095868] vgadash: loading out-of-tree module taints kernel.
[ 2.097549] vgadash: module verification failed: signature and/or required ke
[ 2.104668] printk: console [vgadash-1] enabled
[ 2.105033] vgadash: SysRq toggle enabled (Alt+SysRq+v)
[ 2.105103] vgadash: SysRq logs enabled (Alt+SysRq+g)
[ 2.105158] vgadash: SysRq state enabled (Alt+SysRq+y)
[ 2.105201] vgadash: loaded (console-tap logs enabled)
[ 2.121857] demo_probe: probing device 0000:00:03.0
[ 2.279268] demo_probe: mapping BAR
[ 2.319219] tsc: Refined TSC clocksource calibration: 2687.766 MHz
[ 2.319670] clocksource: tsc: mask: 0xffffffffffffffff max_cycles: 0x26be197b
[ 2.319901] clocksource: Switched to clocksource tsc
[ 2.439441] demo_probe: waiting for device ready
[ 2.599387] demo_probe: timeout path entered
[ 2.599562] demo_probe: simulating a driver probe hang for UGADASH evaluation
[ 4.616956] sysrq: Show UGADASH logs page

```

Figure 2. VGADASH logs page showing recent kernel output through the VGA overlay.

```

UGADASH [page:state]
-----
Kernel: 5.15.0-174-generic
Uptime: 5 s
CPUs online: 1
Mem: total 454 MiB free 422 MiB

This CPU task: pid=17 comm=kworker/0:1

Controls:
Alt+SysRq+v toggle overlay
Alt+SysRq+g show logs page
Alt+SysRq+y show state page

```

Figure 3. VGADASH state page with compact kernel and machine context plus SysRq controls.

## 5.2 Reproducibility

The implementation used for this study is available at <https://github.com/adityagiri3600/vgadash>. This matters primarily for reproducibility: the evaluation can be repeated from the same code base and with the same QEMU-oriented setup used for the examples discussed here.

## 5.3 Controlled Probe-Hang Case Study

Consider a developer modifying risky driver-initialization code in QEMU. After the change, the guest no longer reaches a useful shell and appears to stall during boot. Without a local fallback path, this is close to an opaque failure: the machine is no longer practically usable, yet it has not failed in a way that conveniently yields a shell, a clean crash dump, or an obvious next debugging step.

VGADASH changes that situation by keeping the last meaningful kernel activity visible on screen. In the example shown in Figure 2, the logs page preserves the final probe-related messages and makes it clear that initialization advanced through BAR mapping before stalling in the device-ready timeout path. That is enough to turn "the VM hung" into a more useful diagnosis: the problem is likely in the timeout or readiness logic of the probe path, not in unrelated boot components. Figure 3 then shows the complementary state page, which confirms that the machine remains alive enough to respond through the dashboard path. This example captures the paper's central claim in concrete form: VGADASH helps diagnose a system that would otherwise be alive but effectively undiagnosable through normal shell-based interaction.

## 6. Comparison with Existing Tools

VGADASH does not replace established kernel debugging tools; it occupies a different point in the design space. SysRq offers emergency commands but does not provide an integrated visual dashboard [1]. Netconsole exports logs to the network, which is powerful when networking is healthy but fragile when the network stack is itself part of the problem [3]. Pstore and ramoops preserve crash records across resets, but they are fundamentally post-event mechanisms [4]. Kdump captures rich crash-state data at significant setup cost and is aimed at post-mortem analysis [5]. Earlyprintk is valuable during early boot but is limited in scope and does not offer a persistent live dashboard view [6].

Related systems work also helps position the paper. Crash-only software argues that recovery-oriented systems should reduce the amount of state and machinery required to regain useful operation after failures [8]. That idea aligns with VGADASH's narrow fallback scope: preserve a small but dependable visibility path instead of reconstructing a full debugging environment. Otherworld explores application survival across kernel crashes by shifting recovery into a separate operating environment [9], which is conceptually adjacent but aimed at continuity rather than direct local observability. Reverse Debugging of Kernel Failures in Deployed Systems focuses on reconstructing kernel failures after deployment with reversible execution support [10]. In contrast, VGADASH is intentionally simpler and more immediate: it surfaces live failure-time information on the local screen rather than enabling post-hoc replay or heavyweight crash recovery.

VGADASH is positioned for a narrower question: how much useful observability can be retained for a system that is still alive, but no longer comfortably accessible? Its answer is intentionally lightweight. It focuses on local monitor output, keyboard-driven control, and a unified on-screen view of recent logs and compact system state. This positioning should be emphasized as complementary rather than competitive.

*Table 1. Positioning VGADASH against existing kernel debugging approaches.*

| Tool       | Works without userspace | Needs network | On-screen visibility | Best fit                                            |
|------------|-------------------------|---------------|----------------------|-----------------------------------------------------|
| VGADASH    | Yes                     | No            | Yes                  | Alive-but-unusable systems needing local visibility |
| SysRq      | Yes                     | No            | No                   | Emergency commands without integrated display       |
| netconsole | Yes                     | Yes           | No                   | Remote log export                                   |

|                  |        |    |           |                              |
|------------------|--------|----|-----------|------------------------------|
|                  |        |    |           | when networking is healthy   |
| pstore / ramoops | Partly | No | No        | Post-reboot crash evidence   |
| kdump            | Partly | No | No        | Post-mortem crash capture    |
| earlyprintk      | Yes    | No | Sometimes | Very early boot logging only |

## 7. Limitations

The present prototype is evaluated primarily in QEMU, and that choice is intentional rather than incidental. QEMU is a realistic and widely used kernel-development environment, especially for risky boot-time and driver experiments, so it is an appropriate primary target for the current study [7]. The limitation is therefore not simply "it runs in an emulator," but that the evaluation still covers a controlled environment with a stable VGA text path and a host-controlled keyboard interface. Broader platform diversity, alternate display paths, and more varied virtual hardware have not yet been studied.

VGADASH is also scoped as a fallback observability surface, not a replacement for richer logging, tracing, and crash-analysis infrastructure. It is most effective when present before the failure occurs, because that keeps the recovery path independent of post-failure shell access. SysRq likewise assumes an operational enough system state for the relevant input and deferred-work path to run [1]. Hard lockups, disabled SysRq support, or environments without a suitable VGA text path can therefore reduce usefulness. These boundaries do not undermine the paper's contribution, but they define the class of failures for which the current design is intended.

### 7.1 Threats to Validity

The primary threat to validity is environmental bias: the probe-hang case study is a controlled fault model running in QEMU, and it does not represent every class of kernel failure. In particular, it preserves scheduler activity and avoids hard lockup behavior so that the diagnostic path remains usable. Another threat is workload coverage: the evaluation focuses on boot-time and early-initialization failures and does not explore all classes of kernel regressions. Finally, the prototype assumes SysRq is enabled and reachable, which may not hold in locked-down production systems.

## 8. Future Work

The most valuable next step is broader validation. Real-hardware testing would strengthen the claims beyond QEMU, especially for display behavior and keyboard interaction. Additional platform coverage would also clarify where VGA-based fallback observability remains practical and where other low-level display paths may be needed.

Another promising direction is richer state summarization. The current dashboard deliberately stays small, but future revisions could surface more failure-oriented summaries, such as crash-adjacent subsystem health indicators or more compact driver-related status signals. Operator ergonomics can also improve through more refined repeated-use workflows.

Finally, future evaluation could add lightweight metrics around responsiveness, rendering latency, and repeated invocation cost. Such data would not redefine the system, but they would strengthen the paper by complementing the current scenario-based evidence.

## 9. Conclusion

VGADASH addresses a practical systems problem: the loss of observability when a Linux machine is still alive but no longer reachable through normal userspace-oriented debugging paths. By treating visibility as a kernel-resident fallback capability rather than as a userspace service, the system provides a local dashboard that remains aligned with failure-time needs.

The academic contribution of the paper lies in showing that a small combination of low-level mechanisms--VGA text mode, in-kernel log capture, and SysRq-driven control--can be assembled into a useful recovery-oriented observability surface. The present implementation is intentionally limited, but it demonstrates that meaningful local visibility can be preserved under degraded conditions and studied in a reproducible environment.

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